

ABSTRACT

Milk is a vital component of the human diet, particularly for children and the elderly, due to its rich nutritional content. However, in recent years, the intentional adulteration of milk with substances such as water, detergent, starch, and urea has become a significant public health concern. These adulterants are often added to increase volume or mimic the appearance of purity, posing serious health risks to consumers. Traditional laboratory-based methods for detecting these adulterants are often costly, time-consuming, and inaccessible in rural or small-scale settings.

This project proposes a low-cost, portable, and real-time milk adulteration detection system using sensor technology. The system employs a combination of pH sensors, temperature sensors, and gas sensors integrated with an Arduino Uno microcontroller. Each sensor monitors a specific physical or chemical property of milk to detect potential contaminants. The collected data is processed and displayed through a 16x2 LCD screen, indicating whether the milk is pure or adulterated.

The results demonstrate that the proposed solution is efficient, user-friendly, and suitable for use in households, dairy farms, and milk collection centres. By providing rapid feedback and eliminating the need for specialized equipment or training, this sensor-based system represents a significant advancement in ensuring milk safety and protecting public health.

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Chapter 1

Introduction

Milk is one of the most essential food products consumed daily by people of all age groups due to its rich nutritional value. It is a primary source of calcium, protein, and various essential vitamins and minerals. These nutrients play a vital role in maintaining the overall health and development of individuals. Calcium, for instance, is crucial for the formation of bones and teeth, while proteins help in tissue repair and growth. Milk is particularly important for children, pregnant women, and the elderly, as it provides the necessary nutrients that are essential for their growth, development, and bone health. Due to these benefits, milk is often referred to as a “complete food.” Given its importance in human health and nutrition, maintaining the purity and quality of milk is critical to ensure that consumers receive all the nutritional benefits it offers.

However, in recent years, the adulteration of milk has become a serious public health concern, especially in developing countries. Unscrupulous practices such as the addition of water, detergents, urea, starch, and other harmful chemicals to increase volume and shelf life are rampant in the milk supply chain. These adulterants are introduced to increase profits, as they make the milk appear to be of higher volume or freshness, often without changing its appearance or taste. Unfortunately, these substances not only dilute the nutritional content of the milk but can also be harmful to human health.

The addition of water to milk, for example, reduces its nutritional value and may also affect its taste and texture. Detergents and soaps, which are sometimes used to mimic the texture of milk, pose a significant health hazard and can lead to gastrointestinal infections, kidney damage, and severe poisoning. Urea, commonly added to milk to increase its apparent richness, is harmful to the body and can cause kidney damage, especially when consumed over prolonged periods. Other adulterants, such as starch and synthetic milk, can also cause digestive issues, as they do not break down in the body as natural milk does. The consumption of such adulterated milk can lead to long-term health problems, including kidney failure, heart disease, and even cancer in extreme cases. Furthermore, many of these adulterants are not visible to the naked eye, making it difficult for consumers to identify the contamination.

According to reports by the Food Safety and Standards Authority of India (FSSAI), a significant percentage of milk samples collected from various states across the country are

found to be adulterated. These adulterants, often added in small quantities, do not affect the appearance, texture, or taste of the milk, which makes them nearly impossible to detect without specialized testing. As a result, consumers unknowingly consume harmful milk, putting their health at risk. Conventional laboratory techniques such as chromatography, spectrophotometry, and titration methods are often used to detect milk adulteration. While these methods are highly accurate, they are expensive, time-consuming, and require specialized equipment and trained personnel. This makes them impractical for routine or household-level testing, especially in rural areas, where access to sophisticated equipment is limited. In many cases, the time it takes to receive the results makes it impossible to take immediate action to prevent harm from consuming adulterated milk.

In light of these challenges, there is a growing need for a low-cost, portable, and easy-to-use system that can detect milk adulteration in real time, without the need for specialized knowledge or tools. The aim of this project is to design and implement a *Milk Adulteration Detection System* that utilizes basic sensors such as temperature, pH, and gas sensors, integrated with a microcontroller and an LCD display to provide an accessible, affordable solution for testing milk purity. This system is designed to enable users to quickly assess the quality of milk, detect adulterants early, and promote better public health practices.

The system operates on the principle that the presence of specific adulterants in milk results in measurable changes to its physical and chemical properties. For example, the addition of water to milk affects the milk's temperature and density. Pure milk has a distinct temperature range that differs from diluted milk, which can be detected by a temperature sensor. Similarly, the presence of urea or detergent alters the pH of milk. Urea, a common adulterant, raises the pH of milk, while detergents make it more alkaline. By measuring these changes in temperature and pH, the system can accurately detect the presence of such adulterants. Additionally, spoiled milk or certain chemical adulterants, such as synthetic milk, release gases that can be detected using a gas sensor. These gases are typically produced due to chemical reactions between the milk and the adulterants, which change the chemical composition of the milk. By continuously monitoring these parameters, the system can compare the values to predefined thresholds, helping to determine whether the milk is pure or contaminated.

This sensor-based approach offers several advantages. It is a cost-effective solution that can be implemented with simple components, making it affordable for small-scale dairy farmers, milk vendors, and even individual consumers. The system provides real-time analysis, enabling

immediate results and empowering users to make informed decisions. For instance, dairy farmers can use the device to ensure the milk they are collecting is free from adulterants before selling it, while milk vendors can verify the quality of milk purchased from suppliers. Additionally, the system reduces human error, as the readings are automated and displayed on an LCD screen with clear indicators of the milk's quality. This makes the system easy to use, even for individuals with minimal technical expertise.

The integration of sensors with a microcontroller allows for automation of the detection process, reducing the need for manual testing and improving consistency in results. The LCD interface displays user-friendly messages, such as "Pure Milk" or "Adulterated Milk Detected," making the system straightforward to operate. As the device can be operated in various settings- from rural households to urban dairies-it provides a practical solution that addresses both the need for accessibility and affordability in ensuring milk safety.

1.1. Objectives

The primary objective of this project is to design and implement a practical, low-cost, and user-friendly Milk Adulteration Detection System capable of identifying the presence of adulterants in milk in real time. In recent years, milk adulteration has become a serious issue affecting both public health and food safety. Therefore, this system is intended to serve as a reliable and accessible solution that allows both consumers and milk producers to verify the quality and purity of milk before it is consumed or distributed in the market. By utilizing sensor-based technology and embedded systems, the project aims to eliminate the dependency on expensive laboratory testing methods and provide instant results in a simple and understandable manner. This ensures that even individuals with minimal technical knowledge can operate the system effectively.

The specific objectives of this project are as follows:

1. Design a Low-Cost Detection System:

- Develop a cost-effective system that can be easily produced and deployed in rural and urban areas without the need for expensive laboratory equipment. The system should be affordable for everyday consumers, milk vendors, and small dairy farmers.

2. Utilize Basic Sensors for Detection:

- Implement temperature, pH, and gas sensors as the core components of the detection system. These sensors will monitor key physical and chemical

parameters of milk that change in the presence of common adulterants such as water, urea, detergent, and synthetic milk.

3. Develop a Microcontroller-Based System:

- Integrate the sensors with a microcontroller (e.g., Arduino, ESP32) that processes the data from the sensors and outputs the results. The microcontroller will be responsible for collecting sensor data, comparing it with predefined thresholds, and activating the output display.

4. Real-Time Detection and Display of Results:

- Ensure that the system provides real-time analysis and feedback. Once the adulteration is detected, the system should immediately display the result on an LCD screen with clear and simple indicators like “Pure Milk” or “Adulterated Milk Detected.”

5. Ensure High Accuracy and Consistency:

- Design the system to be accurate and consistent in its readings, minimizing the possibility of false positives or false negatives. This requires calibrating the sensors and setting appropriate thresholds for the detection of different adulterants.

6. Promote Public Awareness and Health Safety:

- Provide a means for consumers, milk vendors, and dairy farmers to ensure the purity of the milk they are selling or consuming. This system will help in preventing the sale and consumption of adulterated milk, which poses significant health risks.

7. Enable Easy and Simple Usage:

- Ensure that the system is easy to operate, requiring minimal training or technical knowledge. The interface should be simple, and the results should be displayed in a user-friendly manner that anyone can understand.

8. Build a Portable, Durable Prototype:

- Design and build a portable prototype that can be used in diverse environments, from dairy farms to milk collection centres and even households. The system should be durable enough to withstand regular use in varying conditions.

9. Provide a Scalable Solution for Future Enhancements:

- Create a modular system that can be upgraded in the future by adding additional sensors or improving the software. This could include integrating wireless

communication (e.g., Bluetooth, Wi-Fi) for remote monitoring or incorporating a mobile app for easier access to the results.

10. Contribute to the Larger Goal of Food Safety:

- Support the global movement toward food safety by providing an affordable, scalable, and reliable system that ensures milk safety. The success of this system could inspire further development of similar solutions for other food products prone to adulteration.

1.2. Advancements

1.Integration of Sensor Technologies:

- The use of multiple sensors (temperature, pH, and gas sensors) in conjunction with a microcontroller represents a significant advancement in the detection of milk adulteration. Unlike traditional methods, which often rely on chemical testing or lab-based techniques, this system provides a real-time, portable solution that is cost-effective and easy to deploy. The use of sensors allows for automated monitoring of milk's physical and chemical properties, making it a scalable solution for everyday use.

2.Miniaturization and Portability:

- Advancements in electronics have made it possible to develop portable, lightweight systems that can be easily carried and used in various environments, such as homes, local dairies, or remote villages. This miniaturization, coupled with the affordability of sensors and microcontrollers, ensures that the system can be used by anyone without the need for bulky, expensive equipment. This portability ensures widespread adoption and makes milk adulteration detection accessible to a broad population.

3.Real-Time Detection and Immediate Feedback:

- A major advancement in this project is the ability to provide real-time results. Traditional lab testing methods require time for processing and analysis, making it impossible for consumers or small-scale vendors to detect adulteration immediately. The sensor-based approach enables instant feedback, which allows for immediate action to be taken if the milk is found to be adulterated. This immediate feedback can improve the safety and quality control in the milk supply chain.

4.Affordable and Scalable Solution:

- The project presents a significant step forward in making milk adulteration detection affordable and accessible. By utilizing basic sensors and an open-source microcontroller, the cost of the system is drastically reduced compared to conventional lab testing. Additionally, the modular nature of the design allows for future upgrades, such as adding more sensors or wireless communication capabilities (e.g., Bluetooth or Wi-Fi), thus offering a scalable solution for large-scale implementation in the future.

5.User-Friendly Interface:

- The integration of an LCD display that shows clear and simple results, such as “Pure Milk” or “Adulterated Milk Detected,” makes the system accessible even to individuals with little to no technical expertise. This user-friendly interface ensures that dairy farmers, vendors, and even consumers can easily operate the system and obtain results without the need for specialized training.

1.3. Hardware Configuration

The hardware configuration of the Milk Adulteration Detection System is designed to provide an effective, portable, and reliable solution for detecting adulterants in milk. It consists of a combination of sensors, a microcontroller, and a display unit working together as an integrated system. The sensors play a crucial role in measuring important physical and chemical properties of milk such as temperature, pH, and the presence of gases. Any variation in these parameters from standard values indicates possible adulteration. For example, a change in pH may suggest the presence of urea or detergent, while abnormal temperature or gas levels may indicate dilution, spoilage, or chemical contamination. These sensors continuously monitor the milk sample and generate signals that reflect its quality.

The data obtained from the sensors is processed by a microcontroller, which acts as the brain of the system. It analyses the sensor readings by comparing them with predefined threshold values to determine whether the milk is pure or adulterated. The final result is then displayed on a 16x2 LCD screen in a simple and user-friendly format, such as “Milk is Pure” or “Adulteration Detected.” The entire hardware setup is compact and can be powered using a battery or external supply, making it suitable for use in homes, dairy farms, and rural areas.

This integration of sensing, processing, and display ensures real-time detection, ease of operation, and reliability, making the system a practical solution for milk quality analysis.

Below are a detailed breakdown of the components and their configuration:

1. Microcontroller (Arduino)

The heart of the system is the microcontroller, which serves as the central processing unit for the data collected by the sensors. It is responsible for receiving input signals from the temperature, pH, and gas sensors, converting these signals into digital values, and analyzing them based on predefined threshold limits. By continuously monitoring these sensor readings, the microcontroller determines whether the milk sample meets the standard conditions of purity or shows signs of adulteration.

In addition to data processing, the microcontroller also controls the overall operation of the system, including decision-making and output display. Once the analysis is complete, it sends appropriate commands to the LCD module to display clear messages such as “Milk is Pure” or “Adulteration Detected.” This automation ensures fast, accurate, and reliable results while minimizing human intervention. Thus, the microcontroller plays a crucial role in integrating all components and enabling real-time detection in the system.

It performs the following functions:

- **Data Acquisition:** Collects data from the temperature, pH, and gas sensors.
- **Signal Processing:** Processes the raw data from the sensors and compares it against predefined threshold values to determine whether the milk is adulterated.
- **Display Control:** Controls the LCD display to show the results (“Pure Milk” or “Adulterated Milk Detected”).
- **Power Management:** Manages the power consumption of the system, ensuring efficient operation, especially for portable use.

2. Temperature Sensor (DS18B20)

The temperature sensor is used to measure the milk’s temperature, which is an important parameter in detecting adulteration. The DS18B20 is a precise and reliable digital temperature sensor that provides accurate readings with minimal calibration. In pure milk, the temperature typically remains within a specific range under normal conditions. However, when adulterants such as water are added, the thermal properties of the milk change, causing noticeable

variations in temperature. For instance, dilution with cold or hot water can shift the temperature away from its expected range, indicating possible adulteration.

The DS18B20 sensor continuously monitors the temperature of the milk sample and sends the measured data directly to the microcontroller in digital form. Unlike traditional analog sensors, it offers high accuracy and noise immunity, making it suitable for real-time applications. The microcontroller then compares the measured temperature with predefined threshold values to determine whether the milk is within the acceptable range. Any significant deviation is flagged as a potential sign of adulteration, contributing to the overall decision-making process of the system.

3. pH Sensor

The pH sensor is used to measure the acidity or alkalinity of the milk, which is a key indicator of its purity and quality. Under normal conditions, pure milk has a pH value in the range of 6.5 to 6.7. Any significant deviation from this range suggests the presence of adulterants such as detergents, urea, or other chemical substances. For example, detergents tend to make the milk more alkaline, while certain contaminants may shift the pH toward acidity. Therefore, monitoring the pH level is an effective way to detect chemical adulteration in milk.

The pH sensor generates an analog voltage that corresponds to the hydrogen ion concentration in the milk sample. This signal is sent to the microcontroller, which converts it into a digital value and compares it with predefined threshold limits. If the measured pH value falls outside the acceptable range, the system identifies the milk as adulterated. This continuous monitoring and analysis enable accurate and real-time detection, making the pH sensor an essential component of the overall system.

4. Gas Sensor (MQ135)

The gas sensor is used to detect gases released by certain adulterants or due to spoilage in milk. The MQ135 sensor is sensitive to a variety of gases such as ammonia, carbon dioxide, and other volatile organic compounds that may be produced when milk is contaminated or chemically altered. Adulterants like detergents, urea, or synthetic substances can release characteristic gases, while spoiled milk often emits ammonia and other unpleasant compounds. Detecting these gases helps in identifying adulteration that may not significantly affect temperature or pH levels.

The MQ135 sensor works by sensing changes in the concentration of gases in the surrounding air and converting them into an electrical signal. This signal, usually in analog form, is sent to the microcontroller for processing. The microcontroller then compares the gas levels with predefined threshold values to determine whether the concentration is within safe limits. If the detected gas levels exceed the normal range, the system flags the milk as adulterated. This additional layer of detection improves the accuracy and reliability of the system by identifying chemical contaminants that might otherwise go unnoticed.

5. LCD Display (16x2 LCD)

The LCD display acts as the primary user interface of the Milk Adulteration Detection System, providing clear and immediate feedback about the quality of the tested milk. A 16x2 LCD module is used, which can display two lines of text with up to 16 characters each. This display is connected to the microcontroller, which sends processed data and commands to show the final result of the analysis. The use of an LCD ensures that the system communicates information in a simple and easily understandable manner, even for users with no technical background.

After analysing the sensor data, the microcontroller displays straightforward messages on the LCD such as “Pure Milk” when all parameters fall within acceptable limits, or “Adulterated Milk Detected” when any abnormal readings are observed. This instant visual output allows users to quickly make decisions regarding the usability of the milk. The LCD display enhances the overall usability of the system by making it interactive, user-friendly, and suitable for real-time applications in homes, dairy farms, and milk collection centres.

6. Power Supply

The power supply is a crucial component of the Milk Adulteration Detection System, as it ensures continuous and reliable operation of all hardware components. To make the system portable and convenient for use in different environments, it can be powered using a rechargeable battery such as a 9V battery or a lithium-ion (Li-ion) battery. Alternatively, USB power can be used when portability is not a primary concern, such as in laboratory or indoor settings. This flexibility allows the system to be used in a wide range of locations, including homes, dairy farms, and remote rural areas where access to a stable power source may be limited.

To maintain proper functioning, a power management circuit is incorporated to regulate the voltage supplied to the components. This ensures that the microcontroller, sensors, and LCD display receive a stable and appropriate voltage, preventing damage and improving system reliability. Efficient power management also helps reduce energy consumption, allowing the device to operate for longer durations when powered by a battery. Overall, the power supply design enhances the portability, durability, and practicality of the system.

7. Additional Components

In addition to the primary components, several supporting elements are used to ensure the proper functioning, flexibility, and durability of the Milk Adulteration Detection System. Components such as resistors, capacitors, and potentiometers play an important role in circuit stability and accuracy. Resistors are used to control current flow, capacitors help in filtering noise and stabilizing voltage, and potentiometers are useful for calibrating sensors and adjusting threshold values. These components ensure that the sensor readings are accurate and reliable, which is essential for correct adulteration detection.

For circuit development and testing, a breadboard and jumper wires are used to connect all components without the need for soldering. This makes it easier to modify, debug, and improve the system during the prototyping phase. Additionally, an enclosure or casing is used to house all the components securely. It protects the system from external factors such as dust, moisture, and physical damage, thereby increasing the lifespan and reliability of the device. Together, these additional components support the overall efficiency, safety, and usability of the system.

1.4 Hardware Configuration Overview

The hardware configuration of the Milk Adulteration Detection System is designed by integrating multiple components into a single, efficient setup. The system mainly consists of sensors such as temperature, pH, and gas sensors, which are connected to a microcontroller that acts as the central processing unit. These sensors are responsible for continuously monitoring the physical and chemical properties of the milk sample, while the microcontroller processes the collected data and makes decisions based on predefined conditions. The final output is displayed on an LCD, providing clear and immediate feedback to the user.

The sensors are connected to the microcontroller through appropriate input pins, allowing seamless data transfer. Each sensor plays a specific role in detecting adulteration-temperature sensors identify thermal variations, pH sensors detect changes in acidity or alkalinity, and gas

sensors identify the presence of harmful gases or volatile substances. These sensors generate signals corresponding to the measured parameters, which are then read and processed by the microcontroller in real time.

The system is powered using either a battery or a USB power source, ensuring flexibility and portability. This allows the device to be used in different environments such as homes, dairy farms, and milk collection centres. A regulated power supply ensures stable voltage to all components, which is essential for accurate sensor readings and proper functioning of the system. The portable nature of the setup makes it especially useful in rural or remote areas where advanced laboratory facilities are not available.

During operation, the system continuously monitors sensor readings and compares them with predefined threshold values that represent the characteristics of pure milk. If all parameters fall within the acceptable range, the milk is considered pure; otherwise, it is identified as adulterated. The result is then displayed on the LCD screen in a simple and user-friendly format, enabling quick decision-making. This real-time detection and display mechanism make the system highly effective, reliable, and easy to use in practical scenarios.

Chapter 2

Literature Survey

SL NO	TITLE	DESCRIPTI ON	DEVICE	PROS	CONS
1.	"Milk Adulteratio n Detection using Sensors and Arduino," 2019	This paper discusses the use of low-cost sensors (pH, temperature, and conductivity sensors) integrated with an Arduino microcontroller for detecting adulteration in milk.	Arduino, pH Sensor, Temperature Sensor, Conductivity Sensor	Low-cost, portable, and easy to use; real-time results; can detect multiple adulterants	Limited detection range; affected by environment al factors; requires calibration.
2.	"Detection of Adulteratio n in Milk using pH and Conductivit y Sensors," 2020	This study focuses on the use of pH and conductivity sensors to identify water and detergent adulteration in milk.	pH Sensor, Conductivity Sensor, Arduino	Simple to implement fast detection; low cost.	Limited to detecting only water and detergent; lacks accuracy in detecting other adulterants.
3.	"Smart Milk Adulteratio n Detection System using IoT," 2021	This paper presents an IoT-based solution for detecting milk adulteration by using sensors for temperature, pH, and gas analysis, with remote monitoring capabilities.	IoT sensors (pH, temperature, gas sensors), Arduino, Wi-Fi Module	Real-time data monitoring ; remote alerts; multi-sensor integration .	Relatively high cost; reliance on internet connectivity; complex setup.
4.	"Automatic Detection of Milk Adulterants	This research integrates multiple sensors (temperature, pH,	Temperature Sensor, pH Sensor, Gas	Detects multiple adulterants ; portable;	Limited to specific adulterants; sensors need

	using Sensor Technology," 2022	and gas) to detect common adulterants like urea, detergent, and starch in milk.	Sensor, Microcontroller	automated process.	periodic maintenance .
5.	"Development of Milk Adulteration Detection System using Arduino and Mobile App," 2023	This paper discusses the development of a system that combines Arduino-based sensors for detecting milk adulteration and integrates with a mobile app for user notifications.	Arduino, pH Sensor, Temperature Sensor, Mobile App	Low-cost, easy to use, real-time monitoring , mobile alerts.	Mobile app dependency; limited range of detectable adulterants; requires smartphone for operation.
6.	"Milk Quality Control: A Review of Technologies for Adulteration Detection," 2021	This review article provides a comprehensive overview of various methods for detecting adulteration in milk, including chemical methods, spectrophotometry, and sensor-based approaches.	Spectrophotometer, pH Sensors, Conductivity Sensors	Detailed review of available technologies; provides comparative analysis.	Mostly theoretical; lacks detailed implementation examples.
7.	"Real-time Detection of Milk Adulteration using Low-Cost Sensors," 2020	This study explores a sensor-based approach using affordable sensors to detect water and urea in milk, providing instant results.	pH Sensor, Conductivity Sensor, Microcontroller	Cost-effective; instant detection; user-friendly interface.	Limited to water and urea detection; not suitable for all types of adulterants.
8.	"Detection of Adulterants in Milk: A Review of the Latest Trends," 2022	This review focuses on new sensor technologies and their application in detecting a wide range of milk adulterants, including harmful chemicals and biological agents.	Various sensors (pH, conductivity, gas, biosensors)	Covers a wide range of adulterants ; emphasizes emerging technologies like biosensors.	The technology is still evolving; implementation can be expensive.

2.1. EXISTING METHODS

Milk adulteration detection has been an important area of research due to its direct impact on public health and food safety. Over the years, various techniques have been developed to identify the presence of harmful substances in milk, ranging from simple traditional methods to advanced laboratory-based analytical approaches. These methods are designed to detect changes in the chemical composition, physical properties, or biological characteristics of milk caused by adulterants such as water, urea, detergents, starch, and synthetic chemicals. The choice of method depends on factors such as accuracy, cost, required expertise, and the environment in which testing is performed.

Despite the availability of numerous detection techniques, many of the existing methods are either time-consuming, expensive, or require specialized equipment and trained personnel. This makes them less practical for routine use, especially in rural areas or at the consumer level where quick and easy testing is needed. As a result, there is a growing need to explore and compare these existing methods to understand their advantages and limitations. The following section provides an overview of the commonly used milk adulteration detection techniques.

Following are the existing methods

1. Chemical Testing
2. Spectrophotometry
3. High-Performance Liquid Chromatography (HPLC)
4. Gas Chromatography-Mass Spectrometry (GC-MS)
5. Thin Layer Chromatography (TLC)
6. Voltammetry
7. Fourier Transform Infrared Spectroscopy
8. Raman Spectroscopy
9. Near-Infrared (NIR) Spectroscopy
10. Enzyme-Linked Immunosorbent Assay (ELISA)
11. Enzymatic Assays
12. Paper-Based Microfluidics

2.2. LIMITATIONS OF EXISTING METHODS

1. Dependency on Laboratory Infrastructure

Most existing detection methods require a laboratory environment with advanced instruments such as spectrophotometers or chromatography equipment. This makes them inaccessible for rural farmers, small vendors, and households.

2. High Operational Cost

The cost of setting up and maintaining a full-fledged milk testing laboratory is very high. Even a single test may incur significant expenses due to the need for reagents, calibration, and technical manpower.

3. Time-Consuming Testing

Traditional chemical testing methods can take several hours to produce results. This delay is unsuitable in scenarios where milk needs to be tested quickly before distribution or consumption.

4. Lack of Portability

Many existing systems are bulky and immobile. This prevents on-site testing at milk collection centres, transportation points, or households-critical spots in the supply chain where adulteration often occurs.

5. Non-Real-Time Detection

Many laboratory tests provide results only after sample processing, making them ineffective for real-time decisions such as accepting or rejecting a milk delivery.

6. Requirement of Trained Personnel

Laboratory and chemical testing techniques demand expertise in handling instruments, interpreting chemical reactions, and performing titration-based methods. Untrained users are unable to use such methods reliably.

7. Health and Environmental Hazards

Chemical testing often involves handling acids, indicators, and reagents that may be hazardous to users and harmful to the environment if not disposed of properly.

8. Lack of Automation

Manual testing methods are prone to human error and inconsistency. They do not support digital logging, threshold alerts, or automated decision-making, which are essential in modern dairy operations.

9. No Visual or User-Friendly Output

Most traditional methods do not provide user-friendly outputs. Consumers or non-technical users may find it difficult to understand the quality of milk based on chemical readings or laboratory reports.

2.3. PROBLEM DEFINITION

Milk adulteration remains one of the most pressing and persistent challenges in the food and dairy industry today. As milk is a perishable commodity that requires careful handling, unscrupulous practices are often employed to artificially enhance its volume, preserve it for longer periods, or improve its appearance. This is usually achieved through the addition of harmful substances such as water, starch, urea, soap, detergent, baking soda, and synthetic chemicals. These adulterants not only dilute the nutritional integrity of milk but can also introduce serious health hazards. Repeated or prolonged consumption of adulterated milk has been linked to severe medical issues, including liver dysfunction, gastrointestinal disorders, metabolic imbalance, and in extreme cases, cancer or reproductive harm. This makes the problem not only an issue of quality control but also one of national public health concern.

Traditional detection methods, while scientifically accurate, are largely ineffective when it comes to everyday monitoring due to their inaccessibility. Most detection techniques currently in use involve laboratory-based testing procedures that require significant infrastructure, expensive reagents, and trained professionals. These methods, although reliable in controlled environments, are often slow, costly, and impractical for mass deployment-especially in rural areas or small dairy farms where milk is collected and distributed without any intermediate quality verification. Moreover, due to logistical constraints and lack of awareness, most consumers are unaware that the milk they consume could be compromised.

Adding to the concern is the lack of a standardized, affordable system for testing milk quality in real-time and at the point of sale or collection. Existing commercial solutions for milk testing are often too expensive for individual or community-level use, making them inaccessible to small-scale farmers, vendors, and households. Mobile testing labs and high-end analytical equipment serve their purpose in large processing plants or government testing centres, but they do not solve the problem at the grassroots level, where the risk of adulteration is highest and testing facilities are non-existent. Furthermore, the absence of a portable, user-friendly

detection system leaves a significant portion of the milk supply chain unchecked and vulnerable.

To address these challenges, there is a growing need for a system that is simple, affordable, and capable of delivering immediate results. Such a system should not require any complex procedures or technical expertise, and must be compact enough to be used anywhere—from dairy farms and transport vehicles to retail shops and homes. The development of a sensor-based milk adulteration detection system offers a promising solution to this gap. Basic sensors can detect measurable changes in the chemical and physical properties of milk. For instance, the pH level of milk is a critical indicator that shifts when acidic or alkaline adulterants are added. Similarly, a gas sensor can detect the presence of ammonia or other volatile compounds that are typically released from spoiled or contaminated milk. Temperature readings can help identify improper storage or dilution with cold water, while combinations of sensor data can reveal more complex adulteration patterns.

The system utilizes temperature, pH, and gas sensors integrated with a microcontroller to collect, process, and analyse data in real-time. The results are displayed on an LCD screen, providing an immediate and clear indication of whether the milk sample is pure or adulterated. This kind of real-time detection system can play a pivotal role in ensuring milk safety, empowering consumers and farmers alike to make informed decisions about the quality of the product. It also helps build trust and transparency within the supply chain by reducing dependency on centralized labs and shifting quality control to the point of origin or consumption.

By implementing such a system, this project aims to make milk adulteration detection accessible to all stakeholders in the dairy industry—from producers and distributors to retailers and end consumers. It not only addresses the current shortcomings of traditional testing methods but also introduces a scalable and practical approach that can be adopted nationwide. With the rise of embedded systems and low-cost sensor technologies, the time is right for deploying such innovative tools to improve public health and food safety outcomes in developing and developed regions alike.

Chapter 3

System Components Overview

3.1 List of Components

List of components used in the system:

1. Arduino Uno
2. pH Sensor
3. Temperature Sensor (DS18B20)
4. Gas Sensor (MQ135)
5. 16x2 LCD Display
6. Power Supply (Battery or DC Adapter)
7. Breadboard
8. Jumper Wires

3.2 Components Overview

1. Arduino Uno:

The Arduino Uno is a microcontroller board based on the ATmega328P. It is the brain of the milk adulteration detection system. It serves as the central unit that reads input from the sensors, processes the data, and outputs the results to the display. The board includes 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection for uploading code, a power jack, and a reset button. It operates at 5V and is highly suitable for embedded and real-time systems due to its ease of use, large community support, and open-source nature. In this project, it collects analog data from the pH, temperature, and gas sensors, processes the readings against predefined thresholds, and determines whether the milk is pure or adulterated.

2. pH Sensor:

The pH sensor is used to measure the hydrogen ion concentration in the milk, which reflects its acidity or alkalinity. This sensor typically consists of a pH probe and a signal conditioning circuit that interfaces with the Arduino. The standard pH range for pure milk lies between 6.5

and 6.7. If the measured pH deviates significantly from this range, it may indicate adulteration with acidic or alkaline substances such as urea or detergent. The sensor outputs an analog voltage that varies depending on the pH of the liquid. This voltage is then read by the Arduino and analysed to assess the purity of the milk. It plays a crucial role in detecting chemical adulterants that alter the natural pH balance.

3. Temperature Sensor (DS18B20):

The DS18B20 is a precision integrated-circuit temperature sensor whose output voltage is linearly proportional to the Celsius (Centigrade) temperature. It does not require any external calibration and provides accurate readings, typically within $\pm 0.5^{\circ}\text{C}$ at room temperature. In this system, it is used to monitor the temperature of the milk sample. Significant deviations from expected temperature ranges may suggest chemical reactions occurring due to the presence of adulterants or spoilage. Since adulterants may affect the thermal properties of milk, monitoring temperature helps in cross-validating other sensor outputs. The DS18B20 is connected to one of the Arduino's analog pins, and it outputs 10 mV per degree Celsius.

4. Gas Sensor (MQ135):

The gas sensor used in this project MQ135 is capable of detecting various harmful gases and volatile organic compounds. It operates on the principle of changing resistance when exposed to different gases. These sensors are sensitive to a range of chemicals that could be present in adulterated milk, such as ammonia, alcohol, and synthetic detergents. The sensor outputs an analog signal that varies with the concentration of the detected gas, which the Arduino then reads and analyses. This component is essential for identifying adulterants that release vapours or chemicals undetectable by pH or temperature sensors. It enhances the system's ability to detect synthetic or volatile contaminants.

5. 16x2 LCD Display:

The 16x2 LCD module is used to visually communicate the results of the detection process to the user. It can display two lines of text, each with up to 16 characters. This LCD is typically interfaced with the Arduino using a 4-bit or 8-bit data bus, along with control lines. After processing the sensor data, the Arduino sends appropriate messages to the LCD display, such as "Milk is Pure" or "Adulteration Detected." The display ensures that even users with no technical background can understand the results. It makes the system more user-friendly and accessible in real-world scenarios such as milk collection centres or households.

6. Power Supply (Battery or DC Adapter):

The system requires a stable power source to function. Typically, a 9V battery or a DC adapter is used to supply power to the Arduino and connected components. The Arduino Uno includes an onboard voltage regulator that ensures a steady 5V supply to the sensors and display. Portability is a major advantage of this setup, allowing the device to be used in field conditions without the need for continuous access to a power outlet. A power supply switch or connector is usually included to control or recharge the source.

7. Breadboard:

The breadboard is a reusable platform used to prototype the circuit without soldering. It allows various components like sensors, the Arduino, and the LCD display to be connected temporarily using jumper wires. It facilitates easy modification, testing, and debugging of the circuit. For educational and testing purposes, a breadboard is essential as it supports rapid assembly and changes before finalizing the design onto a PCB.

8. Jumper Wires:

Jumper wires are used to establish electrical connections between the components on the breadboard and the Arduino board. They come in different types: male-to-male, male-to-female, and female-to-female, depending on the pin configurations. These wires make the system modular and allow components to be easily connected or replaced. They also help in organizing the layout on the breadboard during prototyping.

Chapter 4

METHODOLOGY

4.1. Working of the System:

The milk adulteration detection system is designed to operate as a fully automated solution, utilizing embedded sensors and a microcontroller to offer real-time, cost-effective, and accurate results for adulteration detection in milk. By seamlessly integrating hardware and software components, the system continuously monitors the milk's quality and provides a clear output, aiding in identifying contaminants or adulterants that compromise its safety and nutritional integrity.

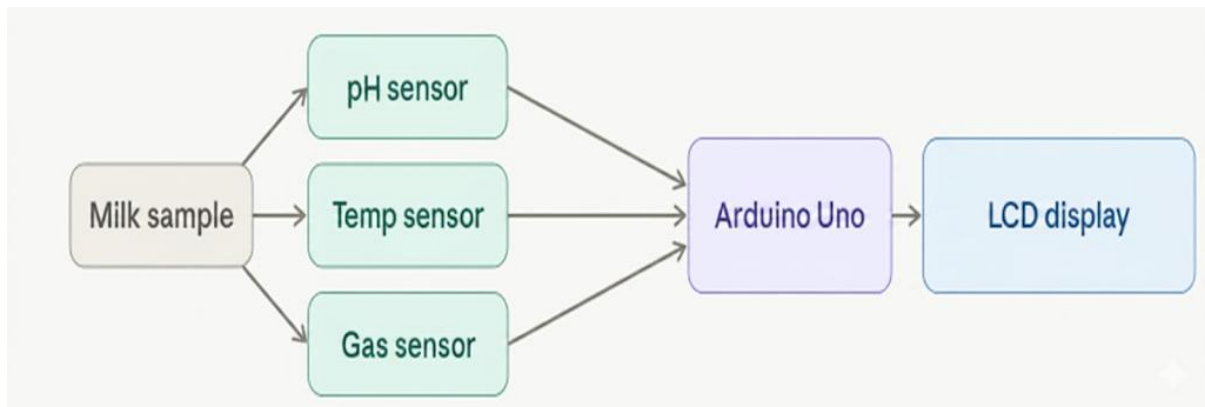


Fig 4.1 Block Diagram

Below is an extensive description of each phase of the system's operation, providing a deeper understanding of its functions and processes:

1. Sample Preparation:

The milk sample preparation is the first step in the adulteration detection process. A small quantity of milk (about 20-30 millilitres) is collected from a milk source, which could range from a milk vendor, dairy farm, or retail outlet. The sample must be free from visible dirt, foreign particles, or excess foam to ensure that the adulteration testing is as accurate as possible. Ideally, the sample should be handled under hygienic conditions to avoid external contamination. Once the sample is prepared, it is placed beneath the sensors or inside a testing chamber that holds the milk steady during analysis. The chamber is specifically designed to

ensure that the sample comes into direct contact with all sensors, guaranteeing a more accurate and thorough reading of the milk's properties.

2. Sensor Activation and Monitoring:

Upon placement of the sample, the sensors are activated, and they begin monitoring various physical and chemical properties of the milk.

- **pH Sensor:** The pH sensor is specifically designed to measure the hydrogen ion concentration in the milk. Normal, unadulterated milk has a pH in the range of 6.5 to 6.7. When adulterants like urea, detergent, or other chemicals are introduced to the milk, it alters the milk's pH value, making it either more acidic or alkaline. The pH sensor continuously tracks these changes and sends real-time data to the microcontroller.
- **Temperature Sensor (DS18B20):** The DS18B20 temperature sensor is employed to monitor the milk's temperature, a critical indicator of adulteration. Pure milk, when freshly collected, should be close to room temperature, generally around 20°C to 25°C. If the milk has been diluted with hot or cold water or subjected to temperature manipulation, the temperature will deviate from this range. This sensor provides real-time temperature readings, which are essential in detecting adulterants such as water or improperly handled milk.
- **Gas Sensor (MQ135):** The gas sensor, typically an MQ35 sensor, detects volatile chemicals or gases released by the milk. Certain chemicals added to milk, such as detergent, urea, or other synthetic compounds, can emit gases like ammonia or other volatile organic compounds (VOCs). These gases are detectable by the gas sensor, which monitors the ambient air around the milk. This is particularly useful in detecting chemical adulterants that might not alter the milk's pH or temperature significantly but release toxic fumes that can be identified with the gas sensor.

3. Data Collection:

As the sensors continuously monitor the milk sample, they generate signals corresponding to changes in pH, temperature, and gas concentration. These signals represent the physical and chemical properties of the milk and are sent to the microcontroller (Arduino Uno) for processing. Since most sensors provide analog outputs, the microcontroller converts these signals into digital values using its internal Analog-to-Digital Converter (ADC). This allows the system to accurately interpret the sensor data and prepare it for further analysis.

The data collection process is carried out continuously to ensure reliable and consistent readings. The pH sensor provides an analog signal indicating the acidity or alkalinity of the milk, the DS18B20 temperature sensor supplies precise temperature data, and the MQ135 gas sensor detects gases released due to adulteration or spoilage. These collected values are then used by the microcontroller to compare with predefined thresholds, forming the basis for determining whether the milk is pure or adulterated.

4. Data Processing and Analysis:

After data collection, the microcontroller (Arduino Uno) processes the sensor data by comparing the readings with predefined threshold values that correspond to the characteristics of pure milk.

- **pH Level Analysis:** The Arduino continuously checks if the pH value falls within the acceptable range of 6.5 - 6.7. If the pH is found to be below this range, it could indicate acidity, often caused by urea or detergent, and the system flags the milk as adulterated. On the other hand, if the pH is above 6.7, the milk could be contaminated with alkaline substances, leading to adulteration.
- **Temperature Analysis:** The microcontroller monitors the temperature readings and checks whether they fall within the normal room temperature range of 20–25°C. Any significant deviation from this range can indicate adulteration with cold or hot water, which is a common method of milk dilution. If the temperature fluctuates beyond acceptable limits, the milk is flagged as adulterated.
- **Gas Detection:** The gas sensor detects the presence of volatile chemicals, such as ammonia, which are commonly associated with detergents or urea. If the sensor detects an abnormal concentration of these gases in the milk's surrounding air, it triggers the system to flag the sample as adulterated.

5. Decision Making and Result Display:

After processing the sensor data, the system makes a decision on the quality of the milk:

- If all the readings (pH, temperature, and gas concentrations) are within acceptable limits, the system concludes that the milk is pure.
- If any of the readings fall outside the expected ranges, indicating a possible adulterant, the system flags the milk as adulterated.

- The result is then displayed on a 16x2 LCD screen. The display shows clear messages such as "Milk is Pure" if all the parameters are within the normal range, or "Adulteration Detected" if any parameter deviates from the standard values.

This real-time output allows the user to make immediate decisions regarding the milk's quality. The system eliminates the need for time-consuming lab tests and provides actionable results in seconds.

6. User Interaction:

The system is designed to be user-friendly, requiring minimal input from the user. Once the milk sample is placed beneath the sensors and the testing process is initiated, the system operates autonomously. The user is not required to make adjustments or manually intervene during the testing process. After the analysis is complete, the result is immediately displayed on the LCD screen. The user can then take action based on the result:

- If the milk is identified as adulterated, the user can discard the sample or report the issue to relevant authorities or vendors.
- If the milk is deemed pure, the user can proceed with confidence, knowing that the milk meets safety standards.
- Additionally, the system could include a warning indicator (such as a red LED or buzzer) that alerts the user immediately when adulteration is detected.

7. Power Supply and Portability:

The system is designed to be highly portable, powered either by a 9V battery or an external adapter, depending on the user's requirements. This makes the system ideal for use in areas without access to mains electricity or where portability is essential (e.g., in rural areas or at the point of sale). The system's low power consumption ensures that it can operate for extended periods on a single battery charge, making it both cost-effective and energy-efficient. In addition, the compact size of the system allows it to be used in various environments, such as dairy farms, milk vendors, or homes, without requiring significant infrastructure or space.

8. Calibration and Maintenance:

To maintain the accuracy of the system, periodic calibration and maintenance of the sensors are essential. Each sensor (pH, temperature, and gas) needs to be calibrated to ensure that the readings remain accurate over time.

- **pH Sensor Calibration:** The pH sensor is calibrated using known pH buffer solutions (e.g., pH 4, 7, and 10) to ensure that the sensor readings are precise and reliable. Calibration should be done regularly to maintain the system's accuracy.
- **Temperature Sensor Calibration:** The DS18B20 temperature sensor is a relatively simple component but may require periodic verification. It can be calibrated against a known temperature standard (e.g., a calibrated thermometer) to ensure accuracy.
- **Gas Sensor Calibration:** The gas sensor must be calibrated against known concentrations of gases (such as ammonia) to maintain its sensitivity and accuracy. This process ensures that the system remains responsive to the presence of volatile chemicals commonly associated with milk adulteration.

Regular maintenance and recalibration of these components help to keep the system in optimal working condition, ensuring that it provides accurate and consistent results.

4.2. Flow chart of the System

The flowchart of the Milk Adulteration Detection System provides a graphical representation of the sequence of operations performed by the system. It explains the step-by-step procedure followed to detect adulteration in milk using sensors and a microcontroller. The flowchart begins with the initialization of the system and proceeds through various stages such as sample collection, sensor activation, data acquisition, processing, and result display. This structured representation helps in understanding the overall functionality and working principle of the system in a simple and organized manner.

Initially, the user collects a milk sample and places it in the testing area. Once the sample is introduced, the pH sensor, temperature sensor, and gas sensor are activated to monitor the physical and chemical properties of the milk. These sensors continuously measure parameters such as acidity, temperature, and the presence of harmful gases or volatile compounds. The collected sensor data is then transmitted to the Arduino Uno microcontroller, where it is converted into digital form and processed for analysis.

After processing the sensor readings, the microcontroller compares the obtained values with predefined threshold limits that represent the characteristics of pure milk. If all the values fall within the acceptable range, the system concludes that the milk is pure and displays the message “Milk is Pure” on the LCD screen. If any parameter exceeds or falls below the threshold limits, the system identifies the presence of adulteration and displays “Adulteration Detected.” Thus,

the flowchart clearly demonstrates the complete operational workflow of the system and highlights its capability for real-time milk quality analysis.

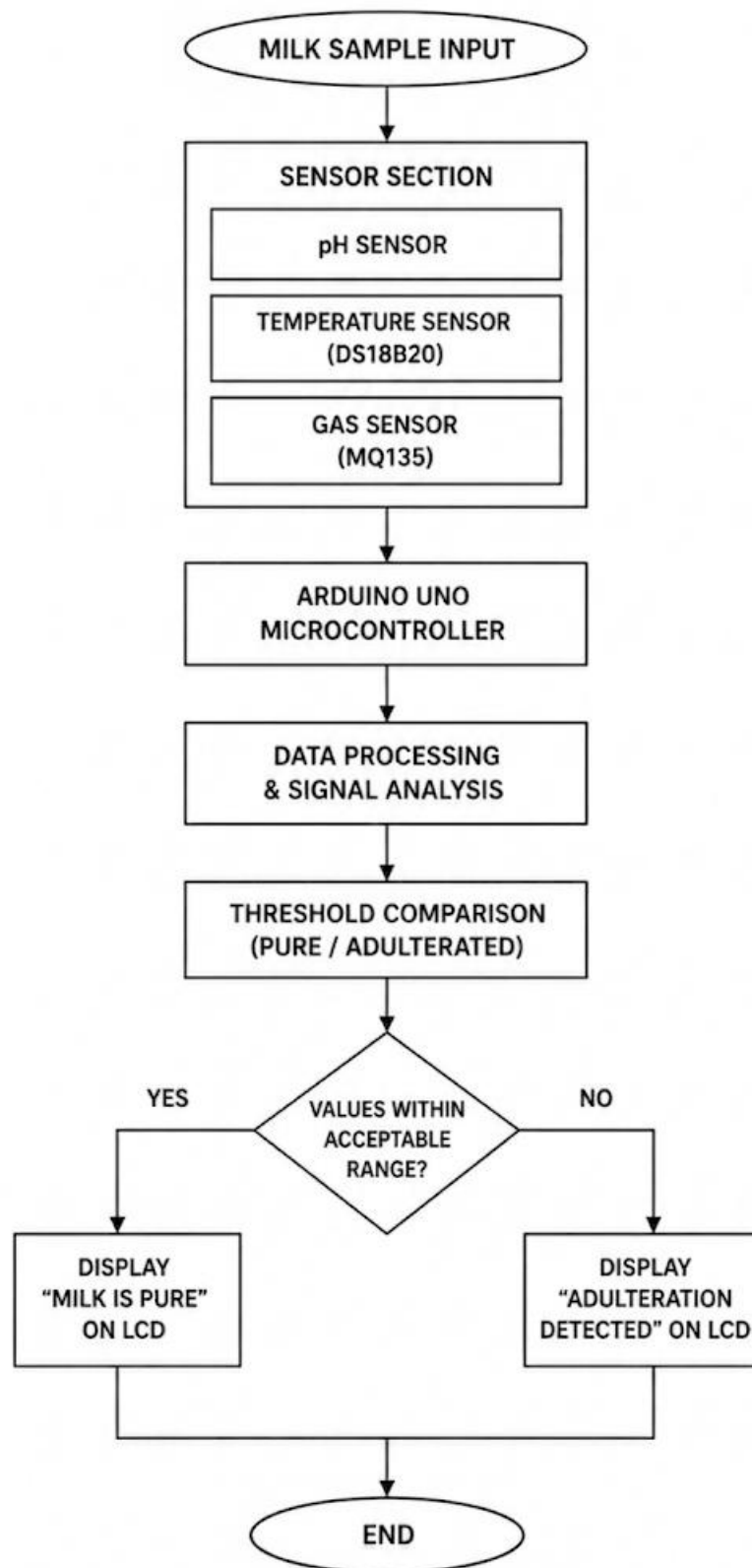


Fig 4.2 Flow chart of the System

4.3. Explanation of the System Flowchart

The flowchart presented visually outlines the operational sequence of the Milk Adulteration Detection System, commencing with system initiation and culminating in the determination and display of the milk's purity status. The process unfolds through the following distinct stages:

1. **Start:** The detection process is initiated upon powering on the system, as indicated by the terminal symbol (rounded rectangle) labelled "Start." This signifies the beginning of the milk quality assessment sequence.
2. **Collect milk sample:** Subsequent to system startup, the user introduces a milk sample for analysis. This manual step, represented by a process symbol (rectangle) labelled "Collect milk sample," provides the input to be evaluated by the sensor array.
3. **Activate sensors:** Following sample collection, the system proceeds to activate the integrated sensor suite, comprising the pH sensor, temperature sensor, and gas sensor. This activation phase, depicted by a process symbol labelled "Activate sensors," prepares these components to acquire relevant data from the milk sample.
4. **Read pH, temperature, gas levels:** Once activated, the sensors perform their respective measurements on the milk sample, quantifying the pH level, temperature, and the concentration of detectable gases. These sensor readings, represented by a process symbol labelled "Read pH, temperature, gas levels," form the raw data for subsequent analysis by the microcontroller.
5. **Values within acceptable range?** This critical decision point, represented by a decision symbol (diamond) labelled "Values within acceptable range?" involves the microcontroller comparing the acquired sensor readings against predefined threshold values established for pure, unadulterated milk.
 - **Yes:** If the measured pH, temperature, and gas levels all fall within their respective pre-established acceptable ranges, the process follows the "Yes" branch, indicating that the milk sample conforms to the characteristics of pure milk based on the measured parameters.

- No: Conversely, if any of the sensor readings deviate from their defined acceptable ranges, the process follows the "No" branch, suggesting potential adulteration or deviation from the expected properties of pure milk.
6. Display "Adulteration Detected": In the event that one or more sensor readings fall outside the acceptable thresholds (following the "No" branch), the system concludes that adulteration is likely present. This outcome is communicated to the user via the LCD display, represented by a process symbol labelled "Display 'Adulteration Detected'."
 7. Display "Milk is Pure": If all sensor readings are within the defined acceptable ranges (following the "Yes" branch), the system determines that the milk sample exhibits characteristics consistent with purity. This result is then displayed to the user on the LCD screen, represented by a process symbol labelled "Display 'Milk is Pure'."
 8. End: Following the display of the result ("Adulteration Detected" or "Milk is Pure"), the current analysis cycle concludes. This termination of the process for the analysed sample is indicated by the terminal symbol (rounded rectangle) labelled "End." To assess a new milk sample, the system would need to be restarted from the "Start" point.

Chapter 5

Results and Discussion

The Milk Adulteration Detection System was successfully designed and implemented using sensors and a microcontroller. The system was tested with different milk samples to evaluate its performance in detecting adulterants. The hardware setup, consisting of the Arduino Uno, pH sensor, temperature sensor (DS18B20), gas sensor (MQ135), and 16x2 LCD display, functioned as expected and provided real-time results.

During testing, the sensors accurately measured the key parameters of milk such as temperature, pH value, and gas concentration. These values were continuously monitored and processed by the microcontroller. Based on the predefined threshold values, the system was able to correctly identify whether the milk sample was pure or adulterated. For instance, when the pH value deviated from the normal range or when abnormal gas levels were detected, the system classified the sample as adulterated.

The LCD display clearly showed the measured values along with the final status of the milk. As seen in the experimental setup, the display provided outputs such as temperature (e.g., 28.1°C), pH value, gas level, and the final result “ADULTERATED.” This confirms that the system is capable of providing instant and easy-to-understand feedback to the user without requiring any technical expertise.

The results demonstrate that the system is efficient, reliable, and suitable for real-time milk quality analysis. It successfully detects variations caused by adulterants and provides accurate output within a short time. Overall, the prototype validates the feasibility of using a sensor-based approach for milk adulteration detection, making it a practical solution for households, dairy farms, and milk collection centres.

5.1. Results

The Milk Adulteration Detection System was successfully tested using different milk samples. The sensors accurately measured pH, temperature, and gas levels, while the Arduino Uno processed the data and displayed the results on the LCD screen. Based on the sensor readings, the system was able to identify whether the milk was pure or adulterated in real time. The results demonstrate that the system is reliable, easy to use, and effective for milk quality analysis.



Fig 5.1(a) LCD displaying results

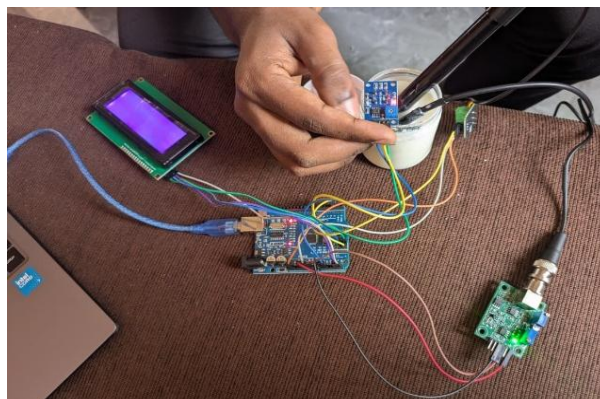


Fig 5.2(b) System Design

5.2. Semester wise Achievements

Sl. No	Semester	Activity Proposed	Detailed Outcome
1	III	Literature Review & Problem Definition	Collected and analyzed existing research papers, technologies, and case studies to identify research gaps and scope for innovation. Finalized the core problem statement, shortlisted innovative and feasible solution ideas
2	IV	Components Study & Software implementation	Studied about the components & implemented the design in the software.
3	V	System Design & Hardware Implementation	Designed system structure, selected components, and planned project timeline and resources. Built and connected hardware components, tested them, and fixed issues to achieve basic functionality.
4	VI	Publication	Prepared documents including abstract, claims, and technical drawings; submitted for publication through official channels.

Chapter 6

CONCLUSION

Ensuring the purity of milk is essential not just for nutritional well-being but also for maintaining public trust in the food supply chain. The widespread problem of milk adulteration poses serious health risks and threatens the integrity of the dairy industry. Substances such as water, detergent, urea, starch, and other chemicals are commonly used to increase quantity or shelf life at the cost of consumer safety. Chronic exposure to such adulterants can lead to diseases ranging from digestive disorders to organ failure and, in some extreme cases, cancer. The need for timely, affordable, and accessible milk testing systems is therefore both a public health necessity and a social responsibility.

The sensor-based milk adulteration detection system proposed in this project offers a practical solution to these challenges. By integrating commonly available sensors with an Arduino Uno microcontroller, the system is capable of detecting a range of adulterants through changes in pH, temperature, and the presence of certain gases. These readings are then analysed and displayed on a simple LCD screen, providing immediate and easy-to-understand results. This approach removes the dependency on centralized labs and highly skilled professionals, enabling anyone—from farmers and dairy vendors to household consumers—to verify milk quality instantly.

The affordability and portability of this system are key strengths. With low-cost components and minimal power consumption, the device is suitable for deployment in remote villages, roadside dairies, school meal programs, and urban households alike. The system can run on battery or a small power source, making it ideal for locations with unreliable electricity. Its minimalistic design ensures that it is not only cost-effective to produce but also easy to maintain and operate.

On a broader scale, the implementation of such systems across the dairy industry could significantly enhance food safety protocols. Integration with cloud-based platforms or mobile apps could allow real-time data transmission to quality control centres, making it possible to identify and act on adulteration trends across regions. For large-scale dairy operations, such real-time monitoring could reduce losses, prevent brand damage, and ensure regulatory compliance.

Moreover, this project opens the door to future improvements and expansions. Additional sensors could be incorporated to detect more sophisticated adulterants, and machine learning algorithms could be implemented to improve detection accuracy over time. The system can be enhanced to provide data logging, report generation, and remote monitoring using IoT modules like Wi-Fi or GSM. These improvements would make it suitable not only for consumer use but also as a regulatory tool for health inspectors and food safety authorities.

Beyond its technical value, the project also serves an educational purpose. It helps students and developers understand how embedded systems and sensor integration can be used to solve real-world problems. It demonstrates how a combination of hardware and software can bring measurable impact to public health initiatives.

In conclusion, the milk adulteration detection system is not just a prototype-it is a model of innovation geared toward community welfare. It combines simplicity, functionality, and affordability in a way that could revolutionize the way we test and trust the milk we consume. As the system evolves with additional features and wider reach, it has the potential to become an essential tool in the global fight against food adulteration, improving the quality of life for millions and promoting a safer, healthier society.

Future Scope

The proposed Milk Adulteration Detection System successfully demonstrates a low-cost and real-time solution for identifying common adulterants in milk. However, with advancements in technology and increasing demands for food safety, there is significant scope for further improvement and expansion of this system. Enhancing the current design can make it more accurate, scalable, and suitable for both domestic and industrial applications.

One of the major areas for future development is the integration of Internet of Things (IoT) technology. By incorporating communication modules such as Wi-Fi, GSM, or Bluetooth, the system can transmit data to cloud platforms or mobile applications. This would enable users to monitor milk quality remotely and store historical data for analysis. Dairy industries and regulatory authorities can use this feature to track milk quality across different stages of the supply chain, ensuring better transparency and control.

Another important improvement lies in the addition of advanced and specialized sensors. While the current system uses pH, temperature, and gas sensors, it can be upgraded by including sensors such as conductivity sensors, optical sensors, and biosensors. These sensors can help detect specific adulterants like starch, sugar, formalin, and antibiotics with higher precision. This enhancement would make the system capable of identifying a broader range of contaminants, thereby improving its effectiveness.

The implementation of machine learning and artificial intelligence (AI) techniques offers another promising future direction. By collecting large datasets from different milk samples, the system can be trained to recognize patterns and predict adulteration more accurately. AI-based models can reduce errors, improve detection efficiency, and adapt to variations in milk quality over time. This would make the system more intelligent and reliable compared to traditional threshold-based methods.

Further scope includes the development of a compact, portable, and user-friendly device. The current prototype can be refined into a more professional product with a durable enclosure, optimized circuit design, and improved power management. A rechargeable battery system with longer backup can be implemented to enhance portability. Additionally, the user interface can be improved through mobile applications or touchscreen displays for better interaction and ease of use.

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Appendix-1

Source Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <OneWire.h>
#include <DallasTemperature.h>

// ----- LCD -----
LiquidCrystal_I2C lcd(0x27, 20, 4);

// ----- PINS -----
#define PH_PIN A0
#define MQ135_PIN A1
#define ONE_WIRE_BUS 2

// ----- TEMP SENSOR -----
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

// ----- CALIBRATION -----
float neutralVoltage = 3.33;
float slope = 3.5;

// ----- STANDARD VALUES -----
#define TEMP_MIN 25
#define TEMP_MAX 35

#define PH_MIN 6.4
```

```
#define PH_MAX 7.0

#define GAS_MAX 300 // only upper limit

void setup() {
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  sensors.begin();

  lcd.setCursor(0,0);
  lcd.print("Milk Quality Sys");
  lcd.setCursor(0,1);
  lcd.print("Initializing...");
  delay(2000);
  lcd.clear();
}

void loop() {

  // ----- TEMPERATURE -----
  sensors.requestTemperatures();
  float temperature = sensors.getTempCByIndex(0);

  // ----- pH SENSOR -----
  float sum = 0;
  for(int i = 0; i < 10; i++) {
    sum += analogRead(PH_PIN);
```



```

    delay(10);
}

float avg = sum / 10.0;
float voltage = avg * (5.0 / 1023.0);

float pHValue = 7 - (voltage - neutralVoltage) * slope;

// ----- MQ135 -----
int gasValue = analogRead(MQ135_PIN);

// ----- STATUS & REASON -----
String status = "PURE MILK";
String reason = "All OK";

if (temperature < TEMP_MIN || temperature > TEMP_MAX) {
    status = "ADULTERATED";
    reason = "Temp Issue";
}
else if (pHValue < PH_MIN || pHValue > PH_MAX) {
    status = "ADULTERATED";
    reason = "pH Issue";
}
else if (gasValue > GAS_MAX) {
    status = "ADULTERATED";
    reason = "Gas High";
}

// ----- LCD DISPLAY -----
lcd.clear();

```

```
lcd.setCursor(0,0);  
lcd.print("T:");  
lcd.print(temperature,1);  
lcd.print("C");
```

```
lcd.setCursor(10,0);  
lcd.print("pH:");  
lcd.print(pHValue,2);
```

```
lcd.setCursor(0,1);  
lcd.print("Gas:");  
lcd.print(gasValue);
```

```
lcd.setCursor(0,2);  
lcd.print("Status:");  
lcd.print(status);
```

```
lcd.setCursor(0,3);  
lcd.print(reason);
```

```
// ----- SERIAL OUTPUT -----
```

```
Serial.println("----- MILK ANALYSIS -----");
```

```
Serial.print("Temperature: ");  
Serial.println(temperature);
```

```
Serial.print("pH Value: ");  
Serial.println(pHValue);
```

```
Serial.print("Gas Value: ");  
Serial.println(gasValue);  
  
Serial.print("Milk Status: ");  
Serial.println(status);  
  
Serial.print("Reason: ");  
Serial.println(reason);  
  
Serial.println("-----\n");  
  
delay(3000);  
}
```